Chapter 14 | Written solutions

Selective State Spaces

All ten exercises are solved here in chapter order. We follow the old state and the new write separately, connect step size to a sigmoid gate, and compare recurrent, dense, and chunked evaluations. The last exercises test causality, an exact discrete reset, and what a memory experiment can actually establish. The written PDF contains every explanation and figure. The notebook PDF also contains the executed code and recorded results.

1. Trace a selective state

The update is $x_t = a_t x_{t-1} + (1 - a_t)u_t$. First multiply the old state by its retention, then add the new input multiplied by its write coefficient. Use the updated state from one row as the old state in the next row.

Position	Input	Old-state part	New-write part	Updated state
1	4	$(1/2)0 = 0$	$(1/2)4 = 2$	2
2	10	$(9/10)2 = 9/5$	$(1/10)10 = 1$	$14/5 = 2.8$
3	2	$(1/4)(14/5) = 7/10$	$(3/4)2 = 3/2$	$11/5 = 2.2$

At position 2, the state moves from 2 to 2.8 toward input 10. It covers $(2.8 - 2)/(10 - 2) = 0.1$ of the distance. This is the write fraction $1 - a_2$. The new-write term itself is 1, while the net change of state is 0.8: the update also discarded 0.2 of the old state. At position 3 the state moves downward because the new input is smaller than the old state.

2. Connect the step to the write

Hold the input u constant over the interval. The equation $\dot{x} = -x + u$ means that the difference $d = x - u$ satisfies $\dot{d} = -d$. After an interval of length Δ, the difference is multiplied by $a = e^{-\Delta}$. Thus $x_{\text{end}} = u + a(x_{\text{start}} - u)$.

Step Δ	Retention a	Write $1 - a$	Control score z
$\ln 2$	$1/2$	$1/2$	0
$\ln 4$	$1/4$	$3/4$	$\ln 3$

The step $\ln 4$ moves the state farther toward the input: three quarters of the starting distance instead of one half. From state 2 toward input 10, the new states are 6 and 8. This distance argument also works when the input lies below the starting state.

To recover the score, solve $\Delta = \log(1 + e^z)$ for $z = \log(e^{\Delta} - 1)$. The sigmoid writes are $\sigma(0) = 1/2$ and $\sigma(\ln 3) = 3/4$. The continuous-time step and the sigmoid gate describe the same restricted scalar update.

3. Respect time order

Write the first map as $f(x) = 0.8x + 2$ and the second as $g(x) = 0.5x - 1$. Substituting the first result into the second gives

$$g(f(x)) = 0.5(0.8x + 2) - 1 = 0.4x.$$

At $x = 3$, the intermediate value is 4.4 and the final value is 1.2. Reversing the order gives

$$f(g(x)) = 0.8(0.5x - 1) + 2 = 0.4x + 1.2.$$

At the same starting value, this result is 2.4. The scalar slopes have the same product in either order, but the later map scales the earlier map's write. That makes the offsets different.

Append $h(x) = 0.25x + 3$. Grouping f and g first gives $h(g(f(x))) = 0.25(0.4x) + 3 = 0.1x + 3$. Grouping g and h first gives $h(g(x)) = 0.125x + 2.75$; then $(h \circ g)(f(x)) = 0.125(0.8x + 2) + 2.75 = 0.1x + 3$. Both groupings return 3.3 at $x = 3$. We changed parentheses without changing which update came first.

4. Follow one input to a read

The write at position s is added after that position's retention has acted on the old state. It crosses transitions $s + 1$ through 3 before the final read $\mathbf{c}_3 = (2, -1)$.

Source s	**Transitions crossed**	**Retention product**	**Read–write product**	$M_{3,s}$
1	At positions 2 and 3	$0.5(0.8) = 0.4$	$2(1) - 1(2) = 0$	0
2	At position 3	0.8	$2(0) - 1(1) = -1$	-0.8
3	None	1	$2(1) - 1(0) = 2$	2

Thus row 3 is $(0, -0.8, 2)$ and $y_3 = -0.8u_2 + 2u_3$ when the initial state is zero. The factor $a_1 = 0.6$ acts on the initial state, so it is absent from every input coefficient in this row. If the initial state were nonzero, its separate contribution would be $a_3 a_2 a_1 \mathbf{c}_3 \mathbf{x}_0$.

A zero output contribution does not imply a zero state contribution. One unit of input 1 leaves the nonzero vector $(0.4, 0.8)^\top$ at position 3. This particular read takes twice the first coordinate and subtracts the second, producing zero. A different read could expose it. The negative and greater-than-one entries also show why these are not softmax probabilities.

5. Inspect a small step numerically

Hold the input fixed during a short interval of length `delta`. In the chapter's scalar zero-order-hold example, the fraction of the old state that survives the interval is $a = e^{-\Delta}$. The coefficient multiplying the input is $b = 1 - e^{-\Delta}$. If the interval is very short, the state barely moves and b is small. Its size is about Δ, since $e^{-\Delta} = 1 - \Delta + \Delta^2/2 - \cdots$ near zero.

Computing `1 - np.exp(-delta)` first rounds the exponential to a floating-point value near one. The subtraction exposes the small difference, so an error of order 10^{-16} in that rounded value can be a large fraction of the write. This loss of significant digits is called **cancellation**. The helper instead uses `-np.expm1(-delta)`. The `expm1` operation evaluates $e^z - 1$ directly and is designed to retain accuracy when z is close to zero.

The reference uses the same binary float64 input values as NumPy. It performs the exponential and subtraction at much higher precision before rounding the small result to float64. The check allows a relative tolerance of 2×10^{-15} and no absolute tolerance, so an erroneous zero would fail even for the smallest positive write.

To measure the loss of precision, divide the absolute error by the reference write for each step. Also try one extra, smaller step to see when subtraction can return zero.

The saved float64 run gives the following write coefficients. The high-precision reference rounds to the stable values shown.

Step	Stable write	Direct subtraction
10^{-4}	$9.999500016666251 \times 10^{-5}$	$9.999500016666385 \times 10^{-5}$
10^{-9}	$9.999999995 \times 10^{-10}$	$9.999999717180685 \times 10^{-10}$
10^{-16}	1.0×10^{-16}	$1.110223024625157 \times 10^{-16}$

For the extra step 10^{-17}, direct subtraction returns zero and the stable write is 10^{-17}.

At 10^{-4}, direct subtraction may lose only the last few digits. At 10^{-9} and 10^{-16}, rounding the exponential can cause a much larger relative error in the small write. For example, returning the float64 gap 1.11022×10^{-16} instead of 10^{-16} is about 11 percent too large. The exact last digits depend on the exponential implementation; the table records this run.

When the rounded exponential is exactly one, subtraction returns zero. The extra step 10^{-17} illustrates that limit on this platform. The stable helper keeps the small positive write even when the retention rounds to one. In exact arithmetic, the two coefficients still sum to one.

6. Give the coordinates different retentions

Now let the state hold two numbers. A row of `a` supplies the retention for each coordinate at that time step. The rows of `b` are all $(1, 1)$, so each new scalar input is written to both coordinates without scaling. The rows of `c` are also $(1, 1)$, so the readout is the sum of the two coordinates after the update.

With separate retentions, the update is

$$x_{t,1} = \tfrac{1}{4}x_{t-1,1} + u_t, \qquad x_{t,2} = \tfrac{3}{4}x_{t-1,2} + u_t, \qquad y_t = x_{t,1} + x_{t,2}.$$

These write coefficients are specified independently of the retention. The rule $b = 1 - a$ in Exercise 5 came from one particular scalar system.

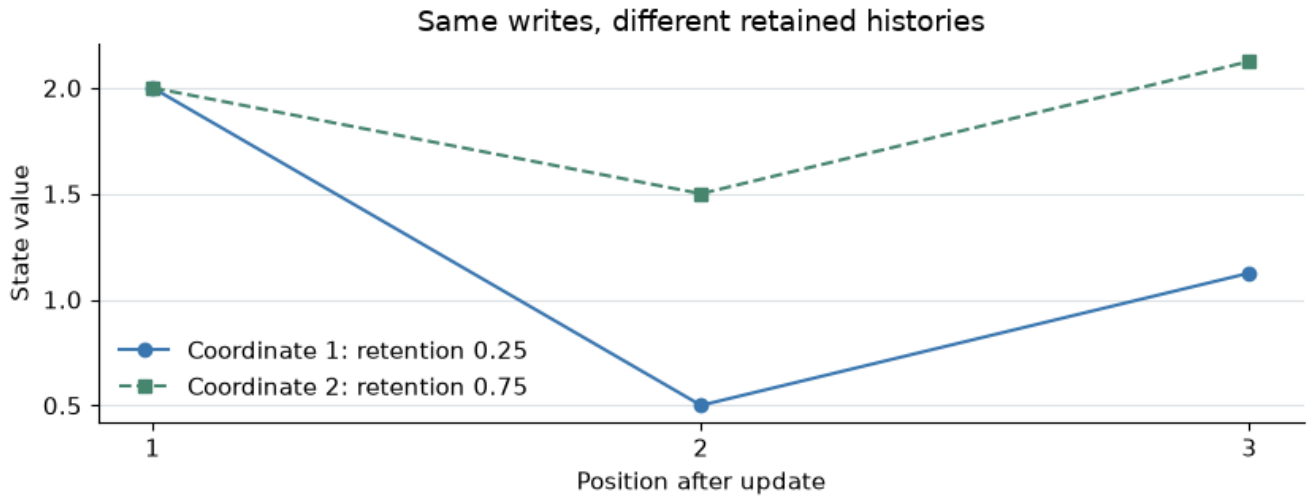
Time t	Input u_t	State $(x_{t,1}, x_{t,2})$	Readout
1	2	$(0 + 2,\ 0 + 2) = (2, 2)$	4
2	0	$(2/4,\ 3(2)/4) = (1/2, 3/2)$	2
3	1	$(1/8 + 1,\ 9/8 + 1) = (9/8, 17/8)$	$13/4$

The second input is zero, which isolates the two retentions. The coordinates enter that step with the same value, 2. The first carries $1/2$ forward and the second carries $3/2$ forward because it retains a larger fraction.

Give `a` shape $[3, 1]$ to store one retention per time step. The scan multiplies a length-one row by the two-coordinate state. NumPy **broadcasts** the length-one row, meaning its single value is used for both multiplications. The state, write, and read widths remain two.

Starting from equal states, the coordinates now receive the same retention and the same write, so they stay equal. After reading 2, each coordinate is 2. After reading 0, each is $2/4 = 1/2$. The final input of 1 gives $(1/4)(1/2) + 1 = 9/8$ per coordinate. The outputs are therefore $(4, 1, 9/4)$. Use absolute tolerance 10^{-12} and zero relative tolerance for these float64 state and output checks.

The first outputs agree because the initial state is zero and both runs apply the same write. The last two outputs in the separate-retention run are each one larger. The first coordinate uses $1/4$ in both runs and contributes no difference. At time 2 the second coordinate differs by $3/2 - 1/2 = 1$. At time 3 its retained contributions differ by $(3/4)(3/2) - (1/4)(1/2) = 1$ before the same new input is added to both. This is a difference in the stored state that the readout exposes.



The zero input at position 2 makes the retention difference visible. Both coordinates started that update at 2, but one falls to 0.5 and the other to 1.5. The final input adds one to both retained values; it does not erase their different histories.

7. Move the chunk boundary

A chunk is a group of consecutive positions processed together. Changing its size moves the boundary where the algorithm must pass its state forward. The sequence and the recurrence are the same for every boundary choice.

Here each state has one coordinate, and each write and read coefficient is one. Starting from $x_0 = 0$, the recurrence is $x_t = a_t x_{t-1} + u_t$ and the output is $y_t = x_t$. The manuscript numbers positions from one. NumPy row 0 therefore refers to position 1 in the calculations below.

Position t	**Retention** a_t	**Input** u_t	**Retained state**	**State and output** $x_t = y_t$
1	$1/2$	1	0	1
2	$1/4$	2	$1/4$	$9/4$
3	$3/4$	0	$27/16$	$27/16$
4	$1/2$	3	$27/32$	$123/32$
5	$1/4$	1	$123/128$	$251/128$

In the dense calculation, row t of M tells us how each input contributes to y_t. An input written at s enters with coefficient one. It passes through the retentions at positions $s+1$ through t before it is read. Thus $M_{t,s} = \prod_{i=s+1}^{t} a_i$ for $s \le t$. The empty product on the diagonal equals one. For $s > t$ the input is still in the future and its coefficient is zero.

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1/4 & 1 & 0 & 0 & 0 \\ 3/16 & 3/4 & 1 & 0 & 0 \\ 3/32 & 3/8 & 1/2 & 1 & 0 \\ 3/128 & 3/32 & 1/8 & 1/4 & 1 \end{bmatrix}.$$

For instance, the input at position 2 passes through $a_3 = 3/4$ and $a_4 = 1/2$ to reach output 4. Its coefficient is $3/8$, which multiplies $u_2 = 2$. Multiplying the whole fourth row by u gives $3/32 + (3/8)2 + (1/2)0 + 3 = 123/32$, exactly the fourth output of the recurrence.

A chunk first computes what its own inputs would produce with a zero incoming state. Then it adds the contribution of the actual incoming state, which is multiplied by every retention from the start of the chunk through the position being read. Those are two different products. A local write skips its own position's retention, while the incoming state has to pass through that retention.

With chunk size 3, the first group covers positions 1–3 and leaves state $x_3 = 27/16$. The group at positions 4–5 is shorter. Its inputs $(3, 1)$ would give $(3, 7/4)$ if this group started from zero. The state from position 3 adds $(27/32, 27/128)$, because it is first multiplied by $1/2$ and then by $1/4$. The sum is $(123/32, 251/128)$. The final state is $251/128$.

Chunk size 1 passes a state after every position. Size 5 contains the whole sequence in one block. Sizes 2 and 3 both need a shorter last block. The assertions below use float64, with an absolute tolerance of 10^{-12} and zero relative tolerance. Compare each output array to both references and the returned state to the final recurrent row of shape `[1]`.

All four chunk sizes agree with the hand calculation. The full sequence matrix displays every possible input–output path in one array. The chunked method evaluates only local blocks directly. It carries the effect of previous blocks in the state at each boundary, which is sufficient for this recurrence.

8. Check causality within a chunk (optional)

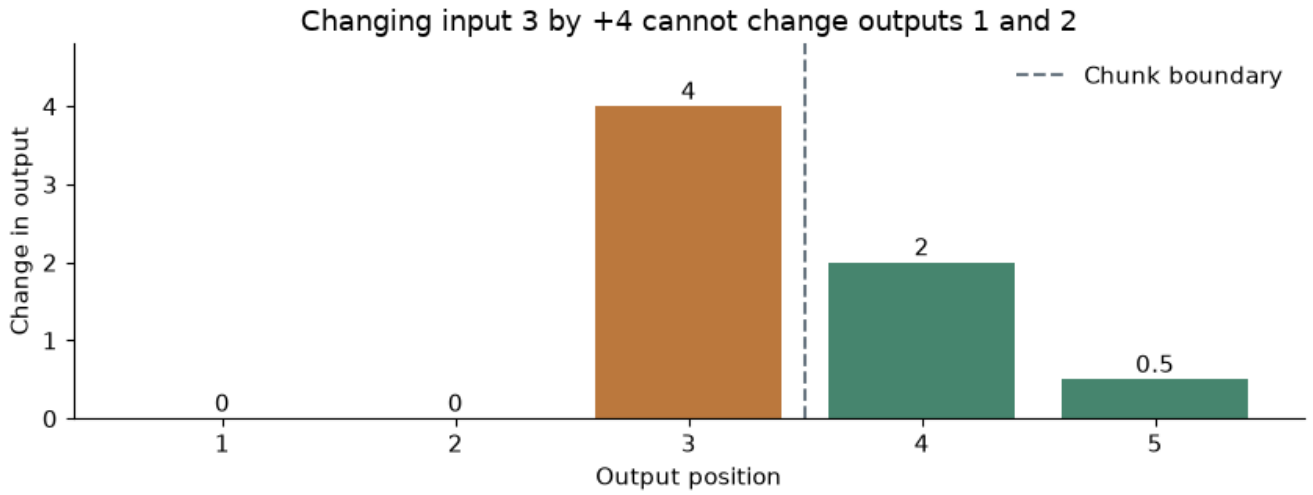
Keep the five-position example from Exercise 7 and use chunks of sizes 3 and 2. Change only the third input, from 0 to 4. The first two inputs now share a chunk with an edited future input. We want their outputs to remain unchanged even though the code computes the chunk together.

The local matrix of the first chunk is the upper-left 3×3 block of M.

$$M_{\text{local}} = \begin{bmatrix} 1 & 0 & 0 \\ 1/4 & 1 & 0 \\ 3/16 & 3/4 & 1 \end{bmatrix}.$$

The first two rows have zeros in column 3. The third row has a one there because the current input is written before its state is read. The change of 4 therefore adds $(0, 0, 4)$ to the first chunk's outputs. The change to its outgoing state is also 4. It passes through retentions $1/2$ and $1/4$ in the next chunk, adding 2 at position 4 and $1/2$ at position 5. The output difference across the full sequence should be $(0, 0, 4, 2, 1/2)$.

The first two outputs remain $(1, 2.25)$. Output 3 increases by 4 and output 4 increases by 2. The triangular local matrix prevents the edit from moving backward within its own chunk, while the state at the boundary carries the edit forward to the next one.



The influence travels only forward. It changes the current output by 4, then crosses the chunk boundary and is reduced by retentions 0.5 and 0.25. The triangle within the first chunk and the state passed to the second chunk enforce the same causal rule.

9. Reset a state inside a chunk (optional)

Use chunk size 3 again. Set $u_3 = 4$ and this time also set $a_3 = 0$. Compare a warm start with $x_0 = 2$ to a cold start with $x_0 = 0$. Since $b_t = c_t = 1$ for every position, the state is again the output. The first two transitions still let the initial state contribute. At position 3, the update is $x_3 = 0 \cdot x_2 + 1 \cdot 4 = 4$ in both runs.

Position t	**Warm output**	**Cold output**	**Warm minus cold**
1	$(1/2)2 + 1 = 2$	1	1
2	$(1/4)2 + 2 = 5/2$	$9/4$	$1/4$
3	4	4	0
4	$(1/2)4 + 3 = 5$	5	0
5	$(1/4)5 + 1 = 9/4$	$9/4$	0

The two runs have the same inputs and coefficients. Their state difference starts at 2 and evolves by $\delta_t = a_t \delta_{t-1}$. Thus the contribution of the initial state to output t is $2 \prod_{i=1}^{t} a_i$. The product contains a zero starting at position 3. This predicts that only outputs 1 and 2 depend on which initial state we choose.

The reset acts on the old state. The new write $b_3 u_3$ is added after that multiplication, so it can still affect position 3 and later outputs. In the matrix, an earlier write must cross a_3 to reach position 3 or beyond, so those coefficients become zero. The coefficient for the current input is $M_{3,3} = 1$. The row of the sequence matrix at position 3 is consequently $(0, 0, 1, 0, 0)$. The first chunk now passes state 4 onward, regardless of which initial state entered it.

The exact reset uses a coefficient supplied directly to the discrete update. For a finite positive Δ in the scalar model, $e^{-\Delta}$ is strictly positive in exact arithmetic. An exact zero is a limiting case, an explicit reset, or a possible floating-point underflow. A merely small retention leaves a merely small dependence on the earlier state.

10. Design two memory tests (optional)

Use the same history with two question variants: the first location and the latest location of a named delivery. For example, let D17 visit North, East, and South, with unrelated D42 reports inserted between them. The correct answers for D17 are North and South. Include different first and last locations so that always returning the latest cannot pass both tasks.

Choose eight location labels with equal tokenized lengths. Randomize IDs and intermediate locations. Sample first and latest labels uniformly as an ordered pair of distinct labels. Keep unrelated labels independent of the target answer. Use a fixed question format and a deterministic answer rule, such as exactly one location label. Reserve whole histories for evaluation; keep both queries and all variants of one history in the same data split.

For a delay comparison, use 8, 32, 128, or 512 reports. Put the first target assignment at report 2, an intermediate update near the middle, and the final target update at report $R - 2$. Fill other positions with neutral reports. Keep the number of target updates and the final-update-to-question distance fixed. Then replace neutral reports with unrelated deliveries or extra target updates in separate comparisons. Match actual token counts and evidence positions after tokenization, since equal report counts need not imply equal sequence lengths.

Report exact-match accuracy for each question, joint accuracy requiring both answers to be correct, and the frequency with which the first-location answer is incorrectly replaced by the latest location. Report these by length and interference level. Uniform guessing among eight labels gives expected accuracy $1/8$ for either individual question.

There is a crucial control: a fixed state can solve both tasks. For one named delivery, store its first location once and overwrite a separate latest-location slot at each target update. Other deliveries leave these slots unchanged. A small oracle demonstrates the algorithm on two histories. The histories share the first location North and latest location South, but their second target locations are East and West, respectively. It tests the task specification, not the performance of a learned SSM.

Both histories leave the oracle's two slots identical, yet their second target locations differ: East versus West. If the query chooses an intermediate update only after the history is processed, these two slots cannot always answer it. For a broader probe, vary the number of target updates and choose a requested update index after the history, with balanced answer labels. Include interior indices and randomized locations. This makes first-and-latest storage insufficient. The example demonstrates a limitation of that particular summary, not a theorem that every finite state fails on every bounded task. If the target ID is also revealed only after the history, the model must retain facts for multiple IDs; report that additional task requirement separately.

Measure persistent inference memory just after the history and before the question. Hold batch size and actual state/cache dtypes fixed. Sum live array elements times their bytes per element across layers. For batch size one and float32, an SSM state of shape $[1, d, N]$ occupies $4dN$ bytes. Add the short convolution buffer and any other persistent history. An attention layer with T cached positions, H_{KV} heads, and key/value widths d_k, d_v stores $4TH_{\mathrm{KV}}(d_k + d_v)$ bytes of key/value payload. Count allocated capacity separately when buffers reserve unused slots. Exclude weights and temporary workspace from this particular persistent-history measurement.

For an illustrative one-layer count, choose $d = 8$, $N = 4$ and a convolution buffer of shape $[1, 8, 3]$. Their float32 payloads total $128 + 96 = 224$ bytes. With two KV heads and key/value widths 4, the attention payload is $64T$ bytes. At $T = 8, 32, 128, 512$, that is $512, 2048, 8192, 32768$ bytes. These are array-size calculations, not measurements of a complete model or evidence about its accuracy.

Keep accuracy and memory results side by side. A fixed-size state can retain the facts demanded by a particular task, while an inadequate learned update may lose them. The extra historical-query probe helps distinguish those possibilities from simply tracking the most recent value.